Linear Algebra and Its Applications

Section 2.1: Matrix Operations

Detailed worked solutions

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Text and numbering. David C. Lay, *Linear Algebra and Its Applications*, fourth edition (Pearson, 2012), as supplied in the textbook PDF linked on Mathjock. Labels give chapter, section, and exercise number; for example, 1.2.3 is Exercise 3 in Section 1.2. A P label denotes a practice problem, and S denotes a chapter supplementary exercise. These are independently written solutions, not a publisher's solutions manual. Short descriptions identify the problems; consult the textbook for the full original statements.

Source scope. The supplied textbook contains printed Chapters 1–8. Its separately published online Chapters 9–10 are absent from that PDF and are not included in this collection.

How to read the solutions. Start with the section's definitions. For a calculation, follow the row operations and substitute the result into the original equations. For a proof, identify the assumptions, the desired conclusion, and why each implication follows. Free variables are parameters, not values that should be silently set to zero. Unless otherwise specified, scalars are real; complex arithmetic is used where the problem requires it.

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2.1 Matrix operations

A matrix has a *shape*: an $m \times n$ matrix has m rows and n columns. Addition requires identical shapes. In AB, the number of columns of A must equal the number of rows of B; the answer then has the outside shape. If A is $m \times n$ and B is $n \times p$, then

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}, \qquad AB = [Ab_1 \ \cdots \ Ab_p].$$

The first formula takes a row against a column; the second says that one column of the answer can be computed without the others. The transpose interchanges rows and columns. In particular $(AB)^T = B^T A^T$: its reversal of order is essential. Matrix multiplication generally does not commute.

Practice problem 2.1.P1: A vector and its transpose

With $A = \begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix}$ and $x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$, first compute $Ax = (5-9, -10+12)^T = (-4, 2)^T$. Thus

$$(Ax)^T = [-4 \ \ 2], \qquad x^T A^T = [5 \ \ 3] \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} = [-4 \ \ 2].$$

The outer product multiplies each entry in the column by the whole row:

$$xx^T = \begin{bmatrix} 25 & 15 \\ 15 & 9 \end{bmatrix}, \qquad x^T x = 5^2 + 3^2 = 34.$$

Finally A^T is 2×2 and x^T is 1×2. The inner sizes $2, 1$ do not agree, so $A^T x^T$ is undefined. An outer product and an inner product are different operations, even when they use the same entries.

Practice problem 2.1.P2: Choose the order of a computation

What we must show. Compute $A^2 x$ efficiently for A of shape 4×4. Associativity permits $A(Ax)$. A matrix–vector product uses four multiplications for each of four entries, or 16 total. Computing Ax then $A(Ax)$ uses $16 + 16 = 32$. In contrast, A^2 has 16 entries, each requiring four multiplications: 64; the final multiplication by x uses 16 more. The second route uses 80. Both routes give the same vector, but the first avoids computing matrix entries that are not needed.

Exercise 2.1.1: Check shapes before calculating

For the matrices supplied in the exercise, A, B are 2×3 and C, D are 2×2. Scalar multiplication and subtraction act entrywise:

$$-2A = \begin{bmatrix} -4 & 0 & 2 \\ -8 & 10 & -4 \end{bmatrix}, \qquad B - 2A = \begin{bmatrix} 7-4 & -5-0 & 1+2 \\ 1-8 & -4+10 & -3-4 \end{bmatrix} = \begin{bmatrix} 3 & -5 & 3 \\ -7 & 6 & -7 \end{bmatrix}.$$

The product AC is undefined: the inner sizes are 3 and 2. For CD, the inner sizes match, and taking rows against columns gives

$$CD = \begin{bmatrix} 1 \cdot 3 + 2(-1) & 1 \cdot 5 + 2 \cdot 4 \\ -2 \cdot 3 + 1(-1) & -2 \cdot 5 + 1 \cdot 4 \end{bmatrix} = \begin{bmatrix} 1 & 13 \\ -7 & -6 \end{bmatrix}.$$

Exercise 2.1.2: Two defined expressions and two shape failures

Entrywise addition gives

$$A+3B=\begin{bmatrix}2+21 & 0-15 & -1+3\\ 4+3 & -5-12 & 2-9\end{bmatrix}=\begin{bmatrix}23 & -15 & 2\\ 7 & -17 & -7\end{bmatrix}.$$

The vector E is 2×1, so $2C-3E$ is undefined: subtraction cannot combine a 2×2 matrix with a 2×1 matrix. For the defined product,

$$DB=\begin{bmatrix}3\cdot 7+5\cdot 1 & 3(-5)+5(-4) & 3\cdot 1+5(-3)\\ -7+4 & 5-16 & -1-12\end{bmatrix}=\begin{bmatrix}26 & -35 & -12\\ -3 & -11 & -13\end{bmatrix}.$$

Finally EC would have inner sizes 1 and 2, so it is undefined.

Exercise 2.1.3: A scalar identity matrix in a sum and a product

For $A=\begin{bmatrix}2 & -5\\ 3 & -2\end{bmatrix}$,

$$3I_2-A=\begin{bmatrix}3-2 & 0-(-5)\\ 0-3 & 3-(-2)\end{bmatrix}=\begin{bmatrix}1 & 5\\ -3 & 5\end{bmatrix},\qquad (3I_2)A=3A=\begin{bmatrix}6 & -15\\ 9 & -6\end{bmatrix}.$$

Subtracting a scalar multiple of I changes diagonal entries only; multiplying by that matrix scales *every* entry.

Exercise 2.1.4: The same distinction in three dimensions

Subtract 5 from each diagonal entry of the supplied A:

$$A-5I_3=\begin{bmatrix}0 & -1 & 3\\ -4 & -2 & -6\\ -3 & 1 & -3\end{bmatrix}.$$

Multiplication instead scales each row, giving

$$(5I_3)A=5A=\begin{bmatrix}25 & -5 & 15\\ -20 & 15 & -30\\ -15 & 5 & 10\end{bmatrix}.$$

Exercise 2.1.5: Compute a product by columns and by entries

(a) Use the two columns of B.

For $A=\begin{bmatrix}-1 & 3\\ 2 & 4\\ 5 & -3\end{bmatrix}$,

$$Ab_1=4\begin{bmatrix}-1\\ 2\\ 5\end{bmatrix}-2\begin{bmatrix}3\\ 4\\ -3\end{bmatrix}=\begin{bmatrix}-10\\ 0\\ 26\end{bmatrix},\qquad Ab_2=-2\begin{bmatrix}-1\\ 2\\ 5\end{bmatrix}+3\begin{bmatrix}3\\ 4\\ -3\end{bmatrix}=\begin{bmatrix}11\\ 8\\ -19\end{bmatrix}.$$

Place them in order: $AB = \begin{bmatrix} -10 & 11 \\ 0 & 8 \\ 26 & -19 \end{bmatrix}$.

(b) Check by the row–column rule.

The six dot products are $-1(4)+3(-2) = -10$, $-1(-2)+3(3) = 11$, $2(4)+4(-2) = 0$, $2(-2)+4(3) = 8$, $5(4)-3(-2) = 26$, and $5(-2)-3(3) = -19$. They give exactly the same matrix, as the two definitions require.

Exercise 2.1.6: A second product, with zeros included

(a) Compute the columns separately.

The columns of A are $(4, -3, 0)^T$ and $(-3, 5, 1)^T$. Hence

$$Ab_1 = \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 12 \\ 3 \end{bmatrix}, \qquad Ab_2 = 4 \begin{bmatrix} 4 \\ -3 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} -3 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 22 \\ -22 \\ -2 \end{bmatrix}.$$

(b) Verify the entries.

The row–column products are $4 - 9 = -5$, $16 + 6 = 22$, $-3 + 15 = 12$, $-12 - 10 = -22$, $0 + 3 = 3$, and $0 - 2 = -2$. Thus $AB = \begin{bmatrix} -5 & 22 \\ 12 & -22 \\ 3 & -2 \end{bmatrix}$.

Exercise 2.1.7: Recover an unknown shape

A 5×3 matrix can multiply B only if B has 3 rows. The product has as many columns as B; since AB has 7 columns, B has shape $\boxed{3 \times 7}$.

Exercise 2.1.8: Rows of a product

The number of rows of BC is the number of rows of B. Therefore B has $\boxed{5}$ rows. The given product does not determine how many columns B has; that number only has to match the rows of C.

Exercise 2.1.9: A parameter making two products equal

Multiplying in the two orders gives

$$AB = \begin{bmatrix} -7 & 3k+18 \\ -4 & k-9 \end{bmatrix}, \qquad BA = \begin{bmatrix} -7 & 12 \\ -k-6 & k-9 \end{bmatrix}.$$

Equality requires equality in *every* position. The upper-right position gives $3k + 18 = 12$, or $k = -2$; the lower-left gives $-4 = -k - 6$, also $k = -2$. The diagonal entries already match. Thus $\boxed{k = -2}$ is both necessary and sufficient.

Exercise 2.1.10: Why cancellation can fail

Direct row–column multiplication gives

$$AB = \begin{bmatrix} -21 & -21 \\ 7 & 7 \end{bmatrix} = AC,$$

although, for example, $B_{11} = -1 \neq -3 = C_{11}$. Indeed $B - C = \begin{bmatrix} 2 & 6 \\ 1 & 3 \end{bmatrix}$ and each column is a multiple of $(2, 1)^T$, which A sends to zero. Thus $A(B - C) = 0$ need not imply $B - C = 0$. Cancelling a matrix requires additional hypotheses.

Exercise 2.1.11: Diagonal multiplication and a commuting example

$$AD = \begin{bmatrix} 5 & 6 & 6 \\ 10 & 12 & 10 \\ 15 & 15 & 12 \end{bmatrix}, \qquad DA = \begin{bmatrix} 5 & 10 & 15 \\ 6 & 12 & 15 \\ 6 & 10 & 12 \end{bmatrix}.$$

The jth column of AD is the jth column of A multiplied by d_{jj}; only the jth entry of the jth column of D is nonzero. Similarly, left multiplication scales row i by d_{ii}. For the last request, choose $B = 2I_3$. This is neither zero nor the identity, and $AB = 2A = BA$ by the identity and scalar-multiplication rules.

Exercise 2.1.12: Build a nonzero matrix annihilated by another

For $A = \begin{bmatrix} 3 & -6 \\ -2 & 4 \end{bmatrix}$, the equation $Ax = 0$ reduces to $x_1 = 2x_2$. Choose two different nonzero solutions, $(2, 1)^T$ and $(4, 2)^T$, and make them columns:

$$B = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}, \qquad AB = [A(2,1)^T \;\; A(4,2)^T] = [0 \;\; 0].$$

Having nonzero columns does not prevent their images from being zero.

Exercise 2.1.13: Pack several matrix–vector products together

Set $R = [r_1 \;\cdots\; r_p]$, an $n \times p$ matrix. The column definition of multiplication gives

$$[Qr_1 \;\cdots\; Qr_p] = QR.$$

The factors are the supplied matrix Q and the matrix of input columns, not auxiliary identity matrices. The sizes are $(m \times n)(n \times p)$. If one of these supplied factors happens itself to be an identity matrix, the literal requirement that neither factor be an identity can still be met: write the product as $(\lambda Q)(\lambda^{-1} R)$. Choose a nonzero scalar λ avoiding any value for which either factor is an identity. Each factor excludes at most one scalar, so at least one of $2, 3, 4$ works.

Exercise 2.1.14: Read the units in a cost product

Column j of Q gives dollars of output of the two products in quarter j. Multiplication by U converts that output vector into three dollar costs: materials, labor, and overhead. Thus UQ is 3×4; its entry (i, j) is the total dollar cost in category i during quarter j. The sum over products is exactly the row–column rule: cost per output dollar times output dollars, added over the two products.

Exercise 2.1.15: Five true–false statements about products and transposes

(a) **False.** Two column vectors a_1, b_1 of shape 2×1 cannot be multiplied as $a_1 b_1$. The correct column formula is $AB = [Ab_1 \;\; Ab_2]$.

(b) **False.** Columns of AB are combinations of columns of A, with weights from columns of B. The roles in the statement are reversed. For example, take $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$. The actual first column of AB is $(1, 2)^T$, whereas the proposed combination using the first column of A as weights gives $b_1 = (1, 1)^T$.

(c) **True.** The left distributive law says $A(B + C) = AB + AC$.

(d) **True.** Entry (i, j) of both $A^T + B^T$ and $(A + B)^T$ is $a_{ji} + b_{ji}$.

(e) **False.** Transposing a product reverses order: $(AB)^T = B^T A^T$. For example, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ give different $B^T A^T$ and $A^T B^T$.

Exercise 2.1.16: Five more structural statements

(a) **True.** Entry j of the first row of AB is $\sum_k a_{1k} b_{kj}$, exactly entry j of $(\text{row}_1 A)B$.

(b) **False.** The proposed expression has one column, whereas AB has three. Its correct form is $[Ab_1 \;\; Ab_2 \;\; Ab_3]$.

(c) **True.** $(A^2)^T = (AA)^T = A^T A^T = (A^T)^2$.

(d) **False.** Two reversals give $(ABC)^T = C^T B^T A^T$. The stated middle order is incorrect; taking $C = I$ reduces the claim to the false same-order product rule in the preceding exercise.

(e) **True.** Transpose distributes over sums, entry by entry.

Exercise 2.1.17: Find two unknown input columns

Write $B = [b_1 \; b_2]$. Since $AB = [Ab_1 \; Ab_2]$, solve the two systems with right sides $(-3, 1)^T$ and $(-11, 17)^T$. For a right side $(c, d)^T$, the equations are $x_1 - 3x_2 = c$ and $-3x_1 + 5x_2 = d$. Replacing the second equation by itself plus three times the first gives $-4x_2 = d + 3c$. For $(c, d) = (-3, 1)$, this gives $x_2 = 2$, $x_1 = 3$; for $(-11, 17)$, it gives $x_2 = 4$, $x_1 = 1$. Thus

$$b_1 = (3, 2)^T, \qquad b_2 = (1, 4)^T, \qquad B = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}.$$

Substitution into both original systems checks the two columns.

Exercise 2.1.18: A zero input column

The third column is Ab_3. If $b_3 = 0$, then $Ab_3 = A0 = 0$. Thus the third column of AB is zero, regardless of the other columns.

Exercise 2.1.19: A sum relation survives multiplication

If $b_3 = b_1 + b_2$, then $Ab_3 = A(b_1 + b_2) = Ab_1 + Ab_2$ by distributivity. Therefore column 3 of AB is the sum of columns 1 and 2.

Exercise 2.1.20: Equal columns have equal images

If $b_1 = b_2$, multiplying the equality by A gives $Ab_1 = Ab_2$. These are precisely the first two columns of AB; they are equal. No conclusion is forced about any later column.

Exercise 2.1.21: A nonzero vector sent to zero

Let b be the last column of B. By hypothesis $b \neq 0$, but the last column of AB is $Ab = 0$. Writing this as a linear combination of the columns of A, the coefficients are the entries of b, not all zero. That is a nontrivial dependence relation. Thus the columns of A are linearly dependent.

Exercise 2.1.22: A dependence relation cannot be destroyed by a linear map

If $b_1, \ldots, b_p$ are dependent, there are scalars c_j, not all zero, with $\sum_j c_j b_j = 0$. Multiply by A and distribute:

$$\sum_j c_j(Ab_j) = A\left(\sum_j c_j b_j\right) = A0 = 0.$$

The *same* nonzero coefficient vector witnesses dependence of the columns Ab_j of AB. Some images may even be zero; that only makes dependence easier, not harder.

Exercise 2.1.23: A left inverse forbids free variables

Suppose $Ax = 0$ and $CA = I_n$. Multiplying by C yields $x = I_n x = (CA)x = C(Ax) = C0 = 0$. Thus the homogeneous system has no free variables: all n columns of A must be pivot columns. Since there is at most one pivot in each row, A must have at least n rows. Therefore it cannot have more columns than rows.

Exercise 2.1.24: Construct a right inverse one column at a time

Since the columns of A span $\mathbb{R}^3$, each equation $Ad_j = e_j$ is consistent for $j = 1, 2, 3$. Choose one solution of each, for example by row reduction, and set $D = [d_1\ d_2\ d_3]$. Each d_j has n entries, so D is

$n \times 3$. Then $AD = [Ad_1\ Ad_2\ Ad_3] = [e_1\ e_2\ e_3] = I_3$. If solutions are nonunique, any choices work; uniqueness was not assumed.

Exercise 2.1.25: A left and right inverse force a square matrix

From $CA = I_n$, the argument of Exercise 23 gives $n \le m$. From $AD = I_m$, every $b \in \mathbb{R}^m$ has solution $x = Db$, so A has a pivot in every row and $m \le n$. Thus $m = n$. Associativity then gives

$$C = CI_m = C(AD) = (CA)D = I_nD = D.$$

One must first respect the sizes; the two inequalities explain why both sorts of inverse can occur together only in the square case.

Exercise 2.1.26: A right inverse guarantees all outputs

For arbitrary $b \in \mathbb{R}^m$, choose $x = Db$. Then $Ax = A(Db) = (AD)b = I_mb = b$. This proves consistency for every right side. Hence A has a pivot in each of its m rows. There cannot be more pivots than columns, so $m \le n$: no more rows than columns.

Exercise 2.1.27: Inner and outer products in coordinates

For $u = (-3, 2, -5)^T$ and $v = (a, b, c)^T$, multiplication gives

$$u^Tv = v^Tu = -3a + 2b - 5c,$$

$$uv^T = \begin{bmatrix} -3a & -3b & -3c \\ 2a & 2b & 2c \\ -5a & -5b & -5c \end{bmatrix}, \qquad vu^T = \begin{bmatrix} -3a & 2a & -5a \\ -3b & 2b & -5b \\ -3c & 2c & -5c \end{bmatrix}.$$

The scalar product pairs corresponding coordinates and adds; the outer product keeps all pairwise products, which explains its different shape.

Exercise 2.1.28: Relationships between the two orders

For real vectors, $u^Tv = \sum_i u_iv_i = \sum_i v_iu_i = v^Tu$. The outer products need not be equal, but $(uv^T)^T = (v^T)^Tu^T = vu^T$. Their (i, j) and (j, i) entries are both u_iv_j. The real-scalar assumption matters: complex inner products usually use a conjugate transpose instead.

Exercise 2.1.29: Prove both distributive laws entrywise

What we must show. Establish Theorem 2(b) and 2(c), with all products of matching sizes. For left distributivity, take an arbitrary entry and expand a finite sum:

$$[A(B + C)]_{ij} = \sum_k a_{ik}(b_{kj} + c_{kj}) = \sum_k a_{ik}b_{kj} + \sum_k a_{ik}c_{kj} = [AB + AC]_{ij}.$$

For right distributivity the same scalar rule gives

$$[(B + C)A]_{ij} = \sum_k (b_{ik} + c_{ik})a_{kj} = [BA + CA]_{ij}.$$

The index k runs over the common inner dimension in each formula. Because every entry agrees, the matrices agree.

Exercise 2.1.30: Move a scalar through a product

For an arbitrary position (i, j),

$$[(rA)B]_{ij} = \sum_k (ra_{ik})b_{kj} = r\sum_k a_{ik}b_{kj} = \sum_k a_{ik}(rb_{kj}) = [A(rB)]_{ij}.$$

The middle quantity is also $[r(AB)]_{ij}$. We used only distributivity of finite sums and associativity of real multiplication. Thus $r(AB) = (rA)B = A(rB)$.

Exercise 2.1.31: The identity on the left

Write $A = [a_1 \ \cdots \ a_n]$, where every $a_j \in \mathbb{R}^m$. By the column definition, $I_m A = [I_m a_1 \ \cdots \ I_m a_n] = [a_1 \ \cdots \ a_n] = A$, using the supplied fact that $I_m x = x$ for each vector in $\mathbb{R}^m$.

Exercise 2.1.32: The identity on the right

The jth column of I_n is e_j, whose only nonzero entry is a 1 in position j. Thus $Ae_j = 0a_1 + \cdots + 1a_j + \cdots + 0a_n = a_j$. Consequently $AI_n = [Ae_1 \ \cdots \ Ae_n] = A$.

Exercise 2.1.33: Prove the reversal rule for transposes

Let A be $m \times n$ and B be $n \times p$. Both $(AB)^T$ and $B^T A^T$ have size $p \times m$. At position (i, j),

$$[(AB)^T]_{ij} = (AB)_{ji} = \sum_{k=1}^{n} a_{jk}b_{ki} = \sum_{k=1}^{n} (B^T)_{ik}(A^T)_{kj} = [B^T A^T]_{ij}.$$

The middle reordering uses commutativity of scalar entries, not commutativity of matrix multiplication. Equality of all entries proves the rule.

Exercise 2.1.34: Transpose a product with a vector

Treat x as a one-column matrix and reverse the order of all three factors:

$$(ABx)^T = x^T(AB)^T = \boxed{x^T B^T A^T}.$$

If A is $m \times n$, B is $n \times p$, and x is $p \times 1$, the final shape is $1 \times m$, as a transposed output vector should have.

Exercise 2.1.35: Create standard matrices in a matrix program

One concrete choice of software is Python with NumPy. These commands create the four requested arrays without entering every entry:

```
import numpy as np
np.zeros((4,5))
np.ones((5,3))
np.eye(5)
np.diag([3,4,2,5])
```

A tuple such as $(4, 5)$ specifies rows first, columns second. The diagonal constructor places the listed numbers on the diagonal and zeros elsewhere. These commands were executed to verify their shapes.

Exercise 2.1.36: Generate reproducible random arrays

The following uses a fixed seed so another reader can repeat the experiment:

```
rng = np.random.default_rng(21036)
R = rng.random((5,6))
Z = rng.integers(-9,10,size=(4,4))
```

Entries of R lie in $[0, 1)$; the endpoint convention differs slightly from the idealized $0 < x < 1$ in the hint and does not affect the task. Entries of Z are integers in $\{-9, -8, \ldots, 9\}$: the upper limit 10 is excluded. The dedicated integer generator avoids having to round scaled real numbers. A seed controls reproducibility, not a mathematical claim that all resulting matrices have a special property.

Exercise 2.1.37: Test commutativity with four random pairs

```
rng = np.random.default_rng(21037)
for j in range(4):
    A = rng.integers(-9,10,(4,4))
    B = rng.integers(-9,10,(4,4))
    print(np.max(np.abs(A@B-B@A)))
```

The executed outputs are $163, 136, 158, 156$. Here $\max|M|$ means the largest absolute entry, not a determinant. Each positive result proves that that particular commutator $AB - BA$ is nonzero, so $AB \neq BA$ in all four trials. One counterexample already disproves a universal commutativity claim. The trials do not prove that matrices *never* commute: I commutes with every square matrix.

Exercise 2.1.38: A valid and an invalid difference-of-squares identity

Use integer arrays to avoid roundoff in testing these polynomial expressions:

```
rng = np.random.default_rng(21038)
for j in range(3):
    A = rng.integers(-9,10,(5,5)); I = np.eye(5,dtype=int)
    print(np.max(np.abs((A+I)@(A-I)-(A@A-I))))
for j in range(3):
    A = rng.integers(-9,10,(4,4))
    B = rng.integers(-9,10,(4,4))
    print(np.max(np.abs((A+B)@(A-B)-(A@A-B@B))))
```

The first three outputs are $0, 0, 0$; the next three are $182, 188, 189$. Algebra explains the difference:

$$(A + I)(A - I) = A^2 - A + A - I = A^2 - I,$$

whereas

$$(A + B)(A - B) = A^2 - B^2 + BA - AB.$$

Thus the latter proposed identity holds precisely when $AB = BA$; the experiments found three pairs for which this extra condition fails.

Exercise 2.1.39: Test three transpose rules

For three random pairs, execute

```
rng = np.random.default_rng(21039)
for j in range(3):
    A = rng.integers(-9,10,(4,4)); B = rng.integers(-9,10,(4,4))
    errors = [(A+B).T-A.T-B.T, (A@B).T-B.T@A.T,
              (A@B).T-A.T@B.T]
    print([np.max(np.abs(E)) for E in errors])
```

The outputs are $(0, 0, 173)$, $(0, 0, 141)$, and $(0, 0, 223)$. Thus the sum rule and reversed-product rule pass all three exact-integer checks, while the same-order product rule fails all three. The proofs in the preceding exercises establish the first two rules for every matching pair; random tests by themselves would not do that.

Exercise 2.1.40: Powers of a shift matrix

The given S satisfies $S(x_1, x_2, x_3, x_4, x_5)^T = (x_2, x_3, x_4, x_5, 0)^T$. Each multiplication shifts entries one more place and inserts another zero. Consequently

$$S^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad S^3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix},$$

$$S^4 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad S^5 = S^6 = 0_{5\times 5}.$$

Equivalently $(S^k)_{ij} = 1$ exactly when $j = i + k$ is a valid column index. Once $k \geq 5$ there is no such position; the matrix is nilpotent.

Exercise 2.1.41: Powers that settle toward an averaging matrix

For the supplied symmetric matrix A, exact rational powers followed by decimal conversion give, for example,

$$A^5 \approx \begin{bmatrix} .333912 & .334877 & .331211 \\ .334877 & .335134 & .329990 \\ .331211 & .329990 & .338799 \end{bmatrix}, \quad A^{10} \approx \begin{bmatrix} .333341 & .333344 & .333315 \\ .333344 & .333350 & .333306 \\ .333315 & .333306 & .333379 \end{bmatrix}.$$

By A^{20} and A^{30} every entry rounds to 0.333333 at six decimal places. The observed behavior is convergence toward the matrix with every entry $1/3$: repeated multiplication increasingly averages the coordinates. For a reproducible numerical calculation, enter the fractions rather than rounded initial entries and use `np.linalg.matrix_power(A,k)`. As an optional explanation using later eigenvalue theory, A has eigenvalues $1, (1+\sqrt{13})/12, (1-\sqrt{13})/12$. The last two have magnitude less than 1 and their powers tend to zero; the surviving eigenspace is the line spanned by $(1,1,1)^T$. Its orthogonal projection is exactly the averaging matrix. This explains the trend rather than extrapolating from a few powers.